Ayush Agarwal

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Summary

Business-focused Data Analyst dedicated to bridging technical data science with actionable corporate strategy. Adept at conducting Root Cause Analysis to solve complex business problems and translating multidimensional data into clear strategic insights. Proven track record of driving ROI through Agile supply chain optimization, demand forecasting engines, and interactive BI dashboards that empower cross-functional decision-making.

Skills

Languages & Databases: SQL (Snowflake, PostgreSQL), Python, Advanced Excel (VBA/Macros)
BI & Visualization: Tableau, PowerBI, Streamlit, matplotlib
Data Platforms & Tools: Databricks, Snowflake, AWS, Git/GitHub, Jupyter
Core Competencies: Root Cause Analysis, Demand Forecasting, Agile Strategy, Dashboards & Reporting

Certifications & Training

- **Snowflake Data Cloud:** Completed foundational technical pathways and earned official data warehousing learner badges.
- **Artificial Intelligence using TensorFlow:** SmartInternz & Google.

Experience

Labcorp Bengaluru, India
Data Analyst Associate Intern Feb 2026 – Aug 2026

- Developed dynamic BI dashboards to visualize complex datasets, translating raw data into actionable insights for non-technical stakeholders.
- Streamlined internal reporting by automating data aggregation workflows, empowering strategic decision-making.
- **Tools Used:** SQL (Snowflake & Databricks), Advanced Excel, PowerBI, Python

Core Projects

QuickBite Express – High-Performance Optimization Engine *Python, Streamlit, Scikit-learn*

- Engineered an intelligent operational optimization engine utilizing machine learning to streamline delivery dispatching and routing decisions.
- Implemented predictive models to accurately estimate demand surges and dynamic constraints, improving overall processing efficiency.

NYC Rental Arbitrage Engine *XGBoost, Tableau, Feature Analytics*

- Executed comprehensive market analysis to pinpoint high-yield financial opportunities, translating algorithmic outputs into business strategies.
- Developed interactive visual models to clearly demonstrate ROI projections and risk distributions across distinct geographic market clusters.

Growth Log – Audience Insight Analytics Dashboard *Tableau/PowerBI, Excel, Performance Metrics*

- Formulated a data-driven content strategy by performing granular audience segmentation and trend analysis on career-guidance metrics.
- Designed specialized analytics dashboards to track key conversion milestones, providing clear visibility into audience retention and market reach.

Research & Publications

Predictive Modelling of Physicochemical Properties (QSPR & QSAR) *Letters in Applied NanoBioScience*

- Authored multiple research manuscripts focused on predicting chemical bioactivity using Topological Indices, deploying ensemble regression models (Random Forest, AdaBoost) to optimize computational accuracy.
- **Published QSPR Paper:** [Link to Paper / DOI: 10.33263/LIANBS143.192]

Urban Plant Suitability Index (UPSI) Framework *Environmental and Sustainability Indicators*

- Designed and computed a novel multi-criteria decision-making framework to evaluate tropical urban ecosystems by synthesizing diverse environmental variables into a single statistical scoring metric.
- **Published Paper:** [Link to Paper]

Education

Vellore Institute of Technology Chennai, India
Bachelor of Technology (B.Tech.) in Your Engineering Branch Sept 2022 – Aug 2026
Cumulative GPA: **8.73/10.0**